Point-Based Weakly Supervised Deep Learning for Semantic Segmentation of Remote Sensing Images

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***Abstract*— Weakly supervised semantic segmentation methods can effectively alleviate the problem of high cost and difficult access to annotation in traditional methods. Among these approaches, point annotated semantic label not only offers a more affordable option but also provides accurate location and category information, playing an indispensable role in current research. However, point annotation labeling encounters challenges such as missing global and texture information, and limiting segmentation accuracy and efficiency while being susceptible to noise interference. For the above problems, a weakly supervised remote sensing image classification framework based on point annotated semantic label is proposed, which consists of three components: data augmentation, Pixel-Net, and iterative superpixel-based sample expansion (ISSE). First, the data augmentation method is used to generate a sufficient number of training samples. Subsequently, the weakly supervised network Pixel-Net is trained using point annotated semantic labels. Pixel-Net incorporates traditional image processing techniques such as edge detection and blurring into deep learning, enabling effective learning of edge and spectral semantic details while reducing the impact of noise on classification results. Finally, ISSE leverages contextual information from superpixels and pseudo-labels to enrich the valuable information in weakly supervised labels, thereby improving the model's classification performance. In the experiments, existing semantic segmentation methods and Pixel-Net are evaluated on the Vaihingen and Zurich Summer datasets, and the effectiveness of ISSE is verified. The results show that Pixel-Net achieves the best segmentation accuracy on both datasets, while ISSE can effectively utilize the existing point annotation labels to mitigate the effect of noise and thus improve the accuracy of weakly supervised semantic segmentation.**

***Index Terms*— Point annotated semantic label, remote sensing, semantic segmentation, weakly supervised learning.**

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I. Introduction

SEMANTIC segmentation of remote sensing images is a crucial task in the field of remote sensing technology, aiming to precisely classify acquired remote sensing image data into different feature classes [1], [2]. Precisely identifying the semantic information of the ground surface in remote sensing images holds significant importance in various fields, including environmental monitoring, urban development planning, and disaster management [3], [4], [5].

In recent years, deep learning techniques have gained significant popularity in various domains, including remote sensing image classification, owing to their exceptional ability to extract deep features [6], [7], [8], [9]. Deep learning models designed for semantic segmentation demonstrate outstanding feature extraction and characterization capabilities, allowing them to learn high-level semantic features directly from raw remote sensing image data. This enables accurate end-to-end classification and effectively enhances the accuracy and efficiency of classification [10], [11], [12], [13], [14], [15]. However, the effectiveness of semantic segmentation approaches heavily relies on a substantial number of semantic labels used to train the models [16]. The cost of producing semantic labels is high, as it requires assigning category information to all pixels in remote sensing images and accurately labeling the edge information of each object. Moreover, because remote sensing images have a wide imaging range, the edges of some objects are blurred and difficult to accurately label, which poses challenges to semantic segmentation methods in large-scale remote sensing image classification, especially in terms of data annotation difficulty.

To reduce the need for manual intervention in training samples, researchers have proposed weakly supervised methods that make use of partially or incompletely labeled data [17]. In the context of remote sensing image classification, weakly supervised learning can reduce the reliance on a large number of fully labeled samples by utilizing alternative forms of annotations such as image-level labels [18], [19], bounding boxes [20], and scribble annotations [21]. Image-level labeling provides information on the categories present in an image without specifying their exact locations. However, this approach may not always meet the accuracy requirements for certain applications. Bounding box labeling involves annotating both the location and category

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information of target instances using rectangular bounding boxes. Nonetheless, applying bounding box labeling to remote sensing images, which are often captured at high altitudes and contain targets with significant scale differences, can be challenging. Scribble annotations typically involve points, lines, or polygons. However, constructing lines and polygons requires more extensive drawing operations, is more subjective to the annotator's interpretation, and proves difficult to apply to large-scale remote sensing image mapping. On the other hand, point annotations offer advantages such as lower production costs, faster annotation speeds, and the ability to provide information about the location and category of corresponding pixels.

In this article, the application of point annotations in semantic segmentation models is discussed and analyzed. Fig. 1(c) illustrates the point annotated semantic label, where only a few pixels in samples are annotated with category information. Experiments demonstrate that point annotations are more cost-effective and convenient to produce compared to fully annotated semantic label. However, there are challenges associated with using point annotated semantic label for deep learning. First, point annotated semantic labels do not provide pixel-level detail, which may lead to misclassification of boundary regions or small targets, and are heavily influenced by noise. Second, the loss of information from point-like samples may hinder the model's ability to differentiate between similar categories or effectively handle complex scenes. To address the above problems, a novel framework for weakly supervised remote sensing image classification based on point annotated semantic labels has been designed. The following are the key contributions of this research.

1) We propose a weakly supervised remote sensing image classification framework based on point annotated semantic label to address the challenges of high sample production costs and difficulties in acquiring samples in remote sensing image semantic segmentation.
2) A lightweight weakly supervised network (Pixel-Net) containing an edge feature extraction module is proposed to capture spectral and edge information in images, which can alleviate the problem of noise effects and edge blurring in weak supervision.
3) An iterative superpixel-based sample expansion (ISSE) is proposed, which can obtain more reliable point samples based on the pseudo-labels obtained and enhance the robustness of the model in complex scenes.
4) The effects of sample size and data augmentation in weakly supervised semantic segmentation based on point labeling are comparatively analyzed, and the effectiveness and applicability of Pixel-Net and ISSE are verified on two datasets.

The remainder of this article is organized as follows. Section II provides a concise review of relevant studies. In Section III, the proposed framework for weakly supervised remote sensing image classification is comprehensively described. The experimental validation of the proposed method is presented in Section IV. Finally, Section V presents concluding remarks to summarize the findings.

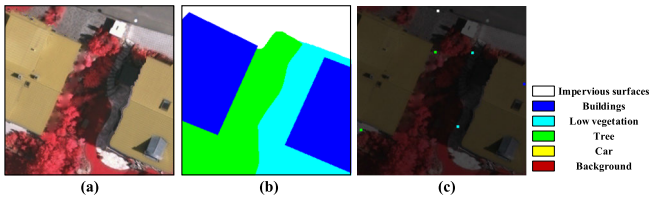


Fig. 1. Illustration of the sample in this article. (a) Images. (b) Fully annotated semantic label. (c) Samples used in this article (point annotated semantic label).

II. Related Work

This section provides a brief review of fully supervised deep learning methods for semantic segmentation of remote sensing images and some weakly supervised learning methods for semantic segmentation of remote sensing images.

Long et al. [10] introduced the full convolutional neural network (FCN), which replaced the fully connected layer with a decoder, achieving end-to-end pixel-level classification and significantly improving prediction efficiency. This led to the development of semantic segmentation networks such as SegNet [22], U-Net [11], and PSPNet [23], widely used in image processing. LinkNet [24] fused deep and shallow features for richer contextual information while retaining high-resolution details. Deeplabv3+ [25] enhanced object and boundary extraction accuracy through null convolution and multiscale feature fusion. HRNetv2 [26] addressed resolution degradation and information loss by introducing high-resolution feature representation and multibranch feature fusion for improved accuracy and detail representation. However, these traditional models require extensive pixel-level labeling, which is laborious and costly.

To address the aforementioned problems, researchers have developed numerous weakly supervised semantic segmentation approaches that utilize coarse or incomplete label information during training. Weakly supervised semantic segmentation labels currently available include image-level labels [27], [28], [29], [30], [31], [32], bounding box-level labels [33], and scribbled annotation labels [34]. Class activation mapping (CAM) can be employed to visually depict feature hotspot maps. CAM is often combined with image-level labeling to generate pixel-level supervised labels representing class hotspots. Xie et al. [35] proposed contrast learning for category agnostic activation map (C2AM) generation without relying on image-level supervision, using only unlabeled image data. Zeng et al. [36] used a saliency map generator instead of traditional CAM, employing superpixels as processing units to obtain more accurate location and edge information. In comparison to image-level labeling-based semantic segmentation methods, Dai et al. [37] introduced Box-Sup, a semantic segmentation approach that solely requires bounding box annotation training, resulting in significant improvements in classification accuracy. Song et al. [38] presented the box-driven class-wise masking model (BCM) for removing irrelevant regions of each class while utilizing filling rate guided adaptive loss (FR-Loss) to aid in disregarding mistakenly labeled pixels in proposals.

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Image-level labels and bounding box-level labels are easy to produce and save a lot of time cost, but they lack pixel-specific annotation information. Scribble annotation labels can not only save the cost of label production but also provide accurate attribute information corresponding to the labeled points [39], [40]. Common scribble annotations can be classified into three categories: point [41], line [34], [42], [43], and polygon [44], [45]. Among them, point annotations have the lowest production cost and can provide precise object locations. Point annotations have attracted a lot of interest from researchers due to their simplicity, efficiency, and effectiveness. Bearman et al. [46] tried to train semantic segmentation models with point labels for the first time and achieved more satisfactory results. Hua et al. [47] complemented the supervised task by considering unsupervised learning signals of neighborhood structure through feature and spatial relation regularization methods. Shin et al. [48] designed an active learning framework called PIXELPICK, which improves weakly supervised accuracy by manually expanding uncertain samples at each iteration. To address problems such as point annotations that do not reveal the complete structure of objects, Wu et al. [49] incorporate bilateral filtering techniques into deep learning, which can help the model better distinguish feature boundaries. Existing weakly supervised methods for remote sensing image classification mainly rely on learning neighborhood information and manual sample expansion to improve classification accuracy, successfully reducing the cost of sample production. However, due to the scarcity and underutilization of labeling information, their classification accuracy is often much lower than that of fully supervised learning. Moreover, the currently available methods do not learn relevant context and texture features sufficiently to accurately learn the detail information in the image, resulting in inaccurate capture of the detailed information in images and high susceptibility to noise interference. Thus, there are still many more aspects that should be investigated in more detail for weakly supervised deep learning-based semantic segmentation of remote sensing images.

III. Methodology

A. Overview

A novel framework for weakly supervised remote sensing image classification is presented in this article, as illustrated in Fig. 2. The proposed framework consists of three parts.

1) *Data Augmentation:* To generate a sufficient number of training samples of size 256 × 256.
2) *Pixel-Net:* Training semantic segmentation models using weakly supervised labeling.
3) *Iterative Superpixel-Based Sample Expansion:* By iteratively expanding more accurate point annotations through the integration of contextual information.

B. Data Augmentation

Data augmentation can improve the model's learning and generalization abilities through a series of transformations and extensions [50], [51], [52], especially in situations where data availability is limited. Traditional fully supervised semantic segmentation and machine learning approaches have demonstrated favorable outcomes with data augmentation techniques. However, the effectiveness of different data augmentation methods in weakly supervised settings remains unclear. In order to investigate the efficacy of various data augmentation methods in weakly supervised scenarios, a comparative analysis experiment is conducted in Section IV-D.

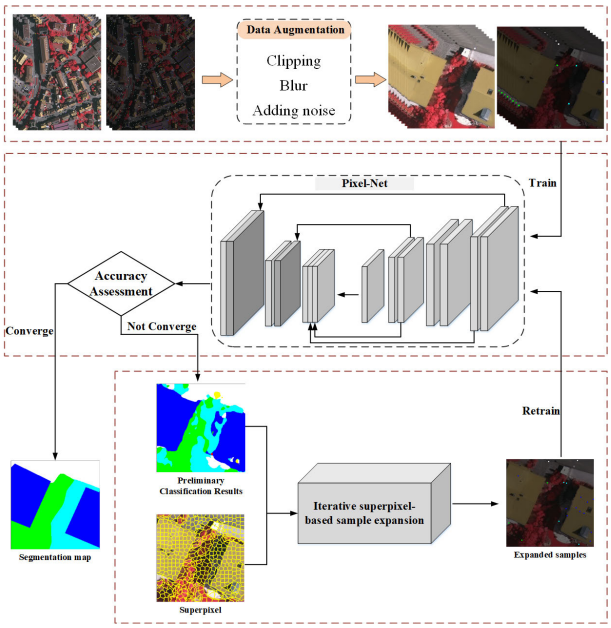


Fig. 2. Overview of the weakly supervised remote sensing image classification framework.

Our experiments reveal that not all data augmentation methods enhance weakly supervised learning models. Among traditional techniques, employing smaller crop stride, blur processing, and adding pretzel noise improves classification accuracy and feature extraction balance. Using a smaller cropping stride not only allows for the reuse of existing point labels but also increases the model's training data. In addition, point annotations from different samples provide varying global information due to context changes. Image blurring technique can be employed to smooth out images and reduce the impact of noise on feature extraction, thus enhancing the model's robustness and its ability to learn global features. Adding salt and pepper noise helps the model build resistance against noise and improves its generalization ability to noisy images. Furthermore, it increases the diversity of data, preventing the model from over-relying on clean data and thus avoiding overfitting to the training data.

C. Pixel-Net

To learn precise target feature representations and reduce the impact of noise, a lightweight weakly supervised model called Pixel-Net is proposed for semantic segmentation of remote sensing images using point annotated semantic label. The structure of Pixel-Net is shown in Fig. 3, and it consists of two main parts: an encoder and a decoder. The output size of the model is 256 × 256 × num_class, and num_class is the

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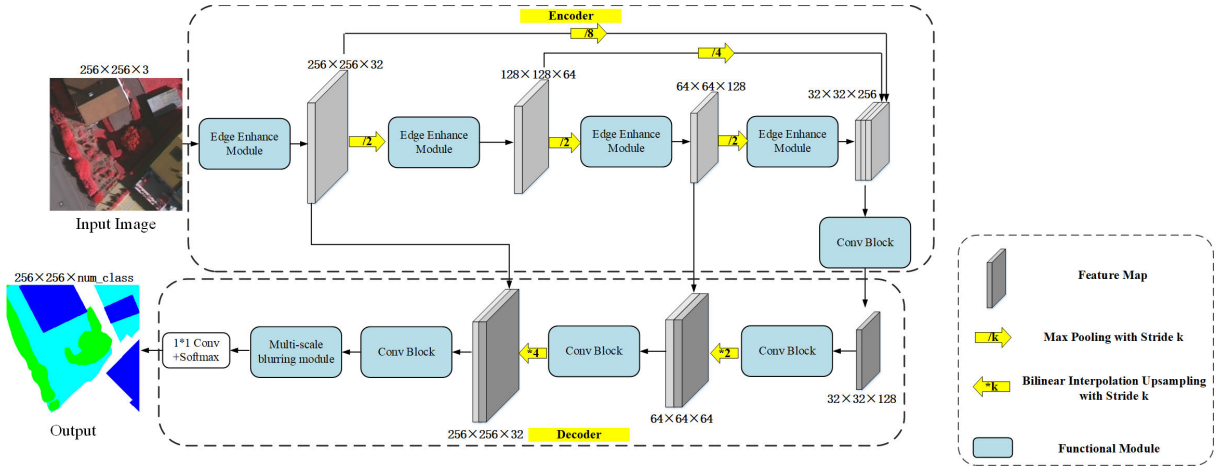


Fig. 3. Structure of the proposed Pixel-Net.

number of categories. The encoder can learn texture features of edges by using the edge enhancement module to extract edge features, while the decoder reduces the impact of noise in the classification results by using the multiscale blurring module. The detailed descriptions of the aforementioned network architecture are provided in Sections III-C1–III-C2.

1) Edge Enhancement Module: The edge enhancement module introduces the Sobel edge operator from traditional image processing methods into deep learning, which is used to extract deep features such as edge texture information and spectral information from remote sensing images. Its structure, as shown in Fig. 4, consists of two feature extraction branches. One branch comprises two 3×3 convolutional layers, which aim to capture the semantic information of the input features. The other branch, known as the edge extraction branch, effectively preserves feature edges and enhances detailed information within the remote sensing image. The edge extraction branch consists of a Sobel convolutional layer, a shortcut connection layer, a 1×1 convolutional layer, and an asymmetric convolutional layer with a kernel size of seven. Specifically, the Sobel convolutional layer consists of two 3×3 convolutional layers with horizontal and vertical Sobel operator weights, and the number of feature channels in the convolutional layer is the same as that of the input features, so the Sobel convolutional layer does not change the number of channels of the features. Moreover, the features generated by these layers highlight the edge information of the input features in different directions. The generated features are then combined by absolute summation to obtain a comprehensive representation of the edges. To further enhance the detail and edge information within the input data, the edge features are added to the inputs, reinforcing fine details and edge features. In addition, a larger convolutional kernel is employed to expand the receptive field of the features and capture more extensive detail information. Finally, the features extracted from the two branches are concatenated together, facilitating the integration of both semantic and edge information.

The two feature extraction branches include convolutional and asymmetric convolutional layers with different scales. Asymmetric convolutional layers can utilize asymmetric

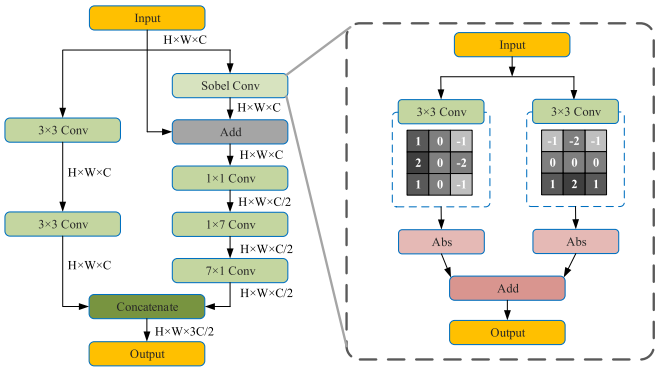


Fig. 4. Structure of the edge enhancement module.

kernels to perform convolution operations, decomposing the traditional $k \times k$ convolutional layer into a $1 \times k$ and a $k \times 1$ convolutional layer from both the horizontal and vertical directions. This can enhance the model's expressive power while controlling the parameter count to $2/k$ of the traditional convolutional layer. Therefore, the edge enhancement module not only enriches the receptive field of the features but also effectively controls the number of parameters in the model compared to the traditional feature extraction structure. Moreover, the incorporation of Sobel convolutional layers applies traditional image processing techniques to deep learning networks, effectively enhancing the edge information in the features. This approach mitigates the problem of blurred edges and inaccurate segmentation results caused by the lack of contextual information in point annotated semantic labels during the training process.

2) Multiscale Blurring Module: To address the issue of severe noise interference in detection results, a multiscale blurring module applies image blurring and multiscale strategies to the end of decoder. The specific structure of this module is illustrated in Fig. 5. The multiscale blurring module comprises four branches, and each branch consists of a max-pooling layer followed by an upsampling layer. To ensure consistent spatial resolution and channel count of the features with the input features, the pooling and upsampling layers have the same step

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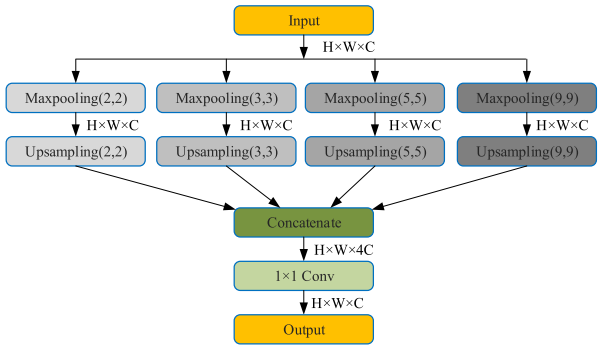


Fig. 5. Structure of the multiscale blurring module.

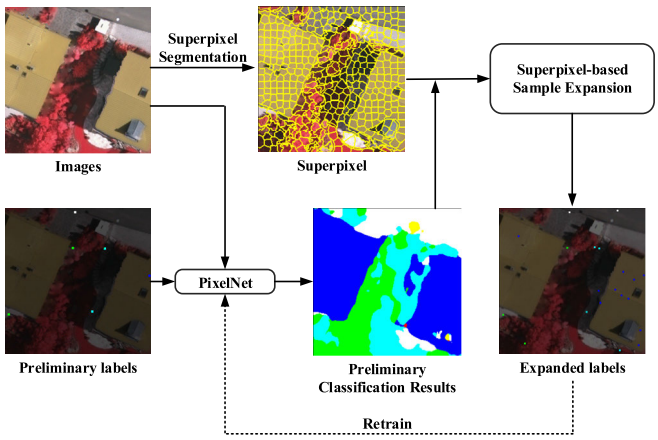


Fig. 6. Structure of ISSE.

size. Based on insights from previous model experiences [23], [53], [54], the step sizes for the four branches are determined as 2, 3, 5, and 9. The features extracted from four branches are subsequently stacked together to create a new feature map. Finally, a 1×1 convolution layer is applied to obtain an output feature map that matches the spatial resolution and the number of channels of the input features.

The downsampling operation plays a role in reducing the image resolution. This process helps to remove high-frequency details and noise while retaining essential structural information. Conversely, the upsampling operation restores the image resolution to its original size or even higher, thus recovering lost details and compensating for information loss during downsampling. Remote sensing images are subject to Gaussian noise and pretzel noise due to the instability of the imaging process through, and by employing upsampling and downsampling layers at different scales, the high-frequency information in the image can be effectively controlled to mitigate the effect of noise at different scales on the classification results. This approach proves to be beneficial in achieving more accurate and robust classification outcomes.

D. Iterative Superpixel-Based Sample Expansion

Weak supervision often leads to suboptimal recognition results due to the lack of accurate annotation information, making it difficult to avoid the impact of noise. ISSE takes superpixels as its analysis object and can effectively expand annotation information by incorporating contextual information.

The ISSE procedure, as shown in Fig. 6, encompasses four main steps. First, the SLIC method is employed for superpixel segmentation of the remotely sensed images. This method eliminates the necessity for predefined segmentation thresholds and offers enhanced processing speed compared to alternative techniques. Subsequently, predictive classification is performed on the images utilizing the existing point annotated semantic labels and the Pixel-Net model. This step entails training the model using the available point annotations and generating pseudo-labels for the unlabeled regions in the image. Next, the labels are expanded through the utilization of the results obtained from superpixel segmentation, the existing point-like labels, and the generated pseudo-labels. This process, referred to as superpixel-based sample expansion (SSE), facilitates the integration of supplementary labeled information from the superpixels into the training set. Finally, the expanded samples are used to retrain the model and generate new predictions, and then, the above steps are repeated until the model accuracy no longer changes.

The SSE algorithm consists of several key steps. First, the algorithm determines the number of additional point annotations required for each sample in this iteration. In this study, the expanded number of point annotations is twice that of the original. Second, a point P is randomly selected from the entire image, and the neighborhood information of the pseudo-label is used to ascertain whether point P lies on the feature's edge. If there are points of different categories in the neighborhood, P is classified as an edge point. Next, the percentage of the predicted category of point P in the superpixel S (which contains point P) is calculated to determine the internal consistency of the superpixel. Third, P is labeled as an extended point sample if it satisfies both the criteria of being a nonedge point and having good internal consistency. Finally, the SSE algorithm expands the number of point annotations by continually iterating through the second and third steps until the predetermined number of point annotations is reached. Through these steps, SSE effectively filters out noise in the edge points and pseudo-labels, thereby enhancing the overall accuracy (OA) of the extended samples and bolstering the robustness and effectiveness of the weakly supervised semantic segmentation task.

IV. Experiment and Analysis

A. Experimental Settings

1) Datasets: They are defined as follows.

a) ISPRS vaihingen dataset: It consists of 33 orthorectified image tile mosaics, each captured with three spectral bands: red, green, and near infrared. In addition, each image in the dataset is accompanied by a corresponding digital surface model (DSM) with a spatial resolution of 9 cm. The dataset encompasses six feature categories, including impervious surfaces, buildings, low vegetation, trees, cars, and background. During the experimentation phase, distinct image IDs were assigned to serve different purposes. Specifically, IDs 11, 15, 28, 30, and 34 were reserved for testing, while

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IDs 1, 10, and 12 were allocated for validation. The remaining 25 images were utilized for training purposes. In addition, the remote sensing image was segmented into image blocks with dimensions of 256 × 256 pixels, employing a clipping step of 256.

b) Zurich summer dataset [55]: It comprises 20 images captured by the QuickBird satellite over the city of Zurich in August 2002. These images have a spatial resolution of 0.62 m and an average size of 1000 × 1150 pixels. Each image consists of four channels: near infrared (NIR), red (R), green (G), and blue (B). In the experimental setup, the R, G, and B channels are exclusively employed. The model is trained on a subset of 15 images, while the remaining images (IDs: 16, 17, 18, 19, and 20) are reserved for testing purposes. The dataset utilized comprises a comprehensive set of urban classes, encompassing road, building, tree, grass, bare soil, water, railway, and swimming pool, resulting in a total of eight distinct classes. Any pixels not assigned to any specific class are labeled as background.

2) Evaluation Metric: Segmentation accuracy was evaluated using two metrics: OA and intersection over union (IoU). OA quantifies the proportion of correctly classified pixels among all annotated pixels, while IoU measures the ratio of the intersection between the predicted and true values of a category to their union. In addition, mean IoU (MIoU) was utilized to assess the model's classification performance for each specific category. Equations are given as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{\mathrm{nc}} \mathrm{TP}_i}{\mathrm{Num}} \tag{1}$$

$$\mathrm{MIoU} = \frac{1}{\mathrm{nc}} \sum_{i=1}^{\mathrm{nc}} \frac{\mathrm{TP}_i}{\mathrm{FN}_i + \mathrm{FP}_i + \mathrm{TP}_i} \tag{2}$$

where TP_i refers to the number of pixels accurately recognized for the ith category, nc is the number of categories, Num is the number of all pixels, FN_i is the number of false predictions for negative samples of the ith category, and FP_i is the number of false predictions for positive samples of the ith category. In the evaluation process, the background category in the Vaihingen dataset consisted of only a small number of insignificant pixels. Therefore, when calculating MIoU, the focus was solely on the other categories, excluding the background. While the Zurich Summer dataset contains a large number of unlabeled pixels, the unlabeled pixels were not considered when calculating the accuracy. Efficiency was also considered by comparing the size of parameters, and the inference time was determined based on the prediction time for unsegmented remote sensing images.

3) Implementation Details: The TensorFlow deep learning framework and a single 2080 GPU with 8-GB memory were used for conducting all experiments. The Adam optimization algorithm was chosen as the optimizer, initialized with a learning rate of 0.001. To prevent overfitting, the learning rate was reduced by half if model validation accuracy failed to improve within four epochs. In addition, all comparison methods are trained for a total of 50 epochs, with a modest batch size of eight. To ensure the effectiveness of the comparative experiment, all methods utilized identical superparameters throughout the study. This article selects categorical cross entropy (CE) as the loss function of the model. Notably, the iteration mentioned in this article refers to the number of cycles of ISSE, not the number of sample inputs in deep learning.

TABLE I

COMPARISON OF THE ACCURACY OF THE EDGE ENHANCEMENT MODULE ON THE VAIHINGEN DATASET USING DIFFERENT EDGE DETECTION OPERATORS

Method	OA	MIoU
Sobel	**79.68%**	**59.66%**
Robert	78.54%	58.01%
prewitt	79.10%	58.76%
Laplacian	78.97%	58.66%

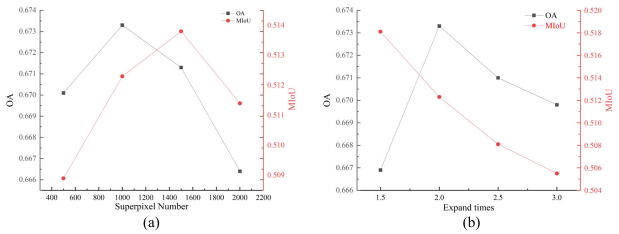


Fig. 7. Effect of superpixel quantity (a) and point annotation expansion multiples (b) on classification performance in the Vaihingen dataset.

4) Parametric Analysis: In order to obtain the edge features in the image and reduce the effect of noise, PixelNet uses the Sobel operator instead of convolution and proposes the ISSE method to expand the number of samples by combining the superpixels. In order to make the model have better performance, this section tests the effects of different edge detection operators, number of superpixels, and point annotations expansion multiplicity on the performance of the model.

Table I shows the accuracy of PixelNet when the edge enhancement module uses different edge extraction operators. It can be seen that the model has the highest classification accuracy when using the Sobel operator. This can be attributed to the small sensory field of the Robert operator, which results in incomplete detection of certain edges. Also, the Laplacian operator cannot capture the image information from different directions. Compared with the Prewitt operator, the Sobel operator can not only extract features from two directions but also consider the distance between neighboring pixels and the center pixel. Therefore, we finally choose to use the Sobel operator to extract the detail information from the image.

Fig. 7(a) shows the influence of the number of superpixels segmented per training sample (256 × 256) on the accuracy in ISSE. Table II presents the time for obtaining different numbers of superpixels for ten training samples. From Fig. 7(a), it can be seen that as the number of superpixels increases, the accuracy shows a tendency to increase and then decrease, reaching optimal accuracy when the number of superpixels is between 1000 and 1500. However, a larger number of superpixels also result in longer segmentation time. Therefore, this study sets the number of superpixels to 1000. The hyperparameter test experiments in this article are conducted

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TABLE II
TIME REQUIRED FOR A DIFFERENT NUMBER OF SUPERPIXELS

Number of superpixels	500	1000	1500	2000
Time(Seconds)	8.2	12.3	16.9	20.8

on the Vaihingen dataset, which can be appropriately adjusted according to the spatial resolution of the remotely sensed images if they need to be applied to other datasets.

Fig. 7(b) shows the effect of the expansion multiplier of the number of point annotations on the accuracy in ISSE. It can be found that with the increase of the expansion multiplier, OA shows a trend of increasing and then decreasing, while MIoU shows a trend of decreasing all the time, which is due to the fact that small features are more difficult to be recognized, and thus, the overfitting problem will occur. Considering the recognition accuracy and the ability to recognize multiple categories of features, we finally expanded the samples to 2× the original sample in each iteration of ISSE.

B. Effect of Sample Size on Weak Supervision

To determine the optimal number of samples needed, the performance of machine learning and semantic segmentation models on the Vaihingen dataset was assessed using different numbers of labels. The classification accuracy of RF in machine learning and Deeplabv3+, HRNetv2, and Pixel-Net in semantic segmentation was examined as the number of labels increased. Specifically, these models were trained using varying numbers of point labels in the training and validation data, and their performance was evaluated on the same test set. The classification accuracy of models utilizing different numbers of point samples was assessed by measuring OA and MIoU.

Fig. 8 illustrates the correlation between the classification accuracy of different methods and the increasing number of point samples. The results indicate that as the number of point-like samples increases, the model's classification performance exhibits an upward trend. Initially, the rate of increase is fast, but it gradually slows down and eventually stabilizes. Notably, when the percentage of point labels was below 0.1%, various models exhibited the most significant improvement in classification accuracy. In addition, the semantic segmentation method demonstrated significantly better classification accuracy compared to the machine learning method. To further evaluate the classification performance of different methods with a small number of point labels, Fig. 9 presents the classification accuracy when the point sample share is below 0.1%. The results highlight that the proposed method in this article outperforms other semantic segmentation methods significantly in terms of classification accuracy when the number of point samples is small.

In general, the model's performance is notably impacted by the increase in the number of point samples when the proportion of such samples falls below 0.01%. Moreover, the cost associated with generating these samples is much lower compared to producing fully annotated semantic labels. As a result, in subsequent studies, the researchers decided to set the

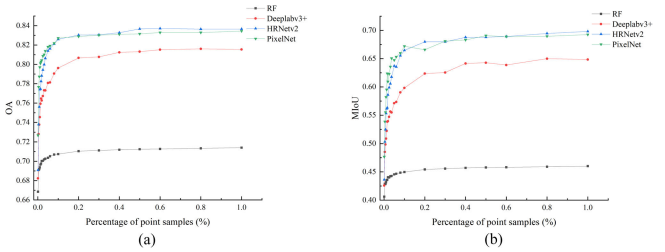


Fig. 8. Impact of point sample quantity on OA and MIoU in the Vaihingen Dataset.

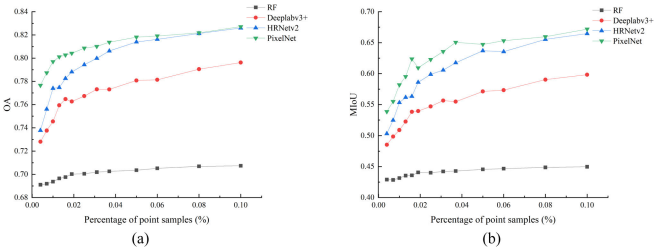


Fig. 9. Impact of point sample quantity on OA (a) and MIoU (b) in the Vaihingen Dataset when the proportion of point annotation is below 0.1%.

TABLE III
NUMBERS OF THE TOTAL NUMBER OF PIXELS AND THE LABELED PIXELS IN POINT-LEVEL AND FULLY SUPERVISED ANNOTATIONS USED IN THIS STUDY

Dataset	Point-level annotations	Fully-supervised annotations	Total number of pixels
Vaihingen	13,041	129,077,878	130,379,957
Zurich Summer	2,005	12,193,952	20,048,660

proportion of point labels in the training dataset to 0.01%. This proportion is equivalent to selecting only approximately six or seven point samples per 256 × 256 image on average. The number of labeled pixels in the point-level and fully supervised annotations used in the training set and the total number of pixels contained in the image are described in Table III.

C. Pixel-Net Evaluation

1) Experimental Comparison and Analysis: To evaluate the performance of the proposed Pixel-Net, contrast experiments are conducted to compare Pixel-Net with the most recent semantic segmentation models and traditional machine learning methods, including support vector machine (SVM) [56], random forest (RF) [57], LinkNet [24], Deeplabv3+ [25], HRNetv2 [26], CRGNet [58], and DBFNet [49] on ISPRS Vaihingen dataset and Zurich Summer dataset [55].

SVM and RF are popular machine learning methods used in various large-scale mapping tasks. SVM is effective at solving linear and nonlinear problems by finding an optimal hyperplane to separate classes of sample points. RF combines stochasticity and multiple decision tree models for predictions using collective voting or averaging. LinkNet, Deeplabv3+, and HRNetv2 are the current state-of-the-art semantic segmentation methods; CRGNet and DBFNet are the latest weakly supervised semantic segmentation methods based on point annotations. LinkNet combines global and local information by connecting feature maps across

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TABLE IV
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE VAIHINGEN DATASET

Method	Supervision	OA	MIoU	Impervious surfaces	Buildings	Low vegetation	Tree	Car
LinkNet	full	80.25%	63.90%	70.57%	81.07%	52.05%	66.89%	48.93%
Deeplabv3+	full	81.82%	64.91%	73.56%	82.64%	55.18%	68.69%	44.48%
HRNetv2	full	83.87%	70.41%	75.88%	84.63%	59.33%	71.05%	61.15%
CRGNet	full	79.18%	59.45%	67.42%	77.31%	51.64%	68.12%	32.77%
DBFNet	full	82.61%	65.53%	73.73%	81.72%	57.55%	70.74%	43.92%
Pixel-Net	full	83.70%	69.79%	75.87%	84.43%	59.14%	70.65%	58.84%
SVM	point	65.87%	39.39%	54.30%	48.78%	38.74%	55.14%	0.00%
RF	point	69.37%	43.15%	56.51%	59.61%	38.92%	58.80%	1.92%
LinkNet	point	73.16%	50.47%	60.56%	69.22%	44.43%	61.01%	17.15%
Deeplabv3+	point	74.54%	50.89%	63.93%	70.02%	45.28%	61.34%	13.88%
HRNetv2	point	77.40%	55.32%	66.57%	74.34%	49.31%	65.26%	21.14%
CRGNet	point	74.99%	48.91%	65.49%	71.38%	43.85%	61.60%	2.24%
DBFNet	point	77.84%	55.71%	66.74%	74.47%	50.47%	66.06%	20.83%
Pixel-Net	point	**79.68%**	**59.66%**	**69.59%**	**78.95%**	**52.17%**	**67.69%**	**29.88%**

TABLE V
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE ZURICH SUMMER DATASET

Method	Supervision	OA	MIoU	Buildings	Grass	Bare soil	Roads	Swimming pools	Trees	Railways	Water
LinkNet	full	77.22%	57.32%	65.75%	70.18%	60.05%	70.08%	55.16%	56.06%	43.21%	38.08%
Deeplabv3+	full	82.76%	61.82%	66.60%	75.78%	57.55%	65.09%	30.29%	74.33%	38.71%	86.20%
HRNetv2	full	90.84%	75.81%	83.17%	83.35%	61.66%	81.84%	43.41%	84.09%	**74.31%**	94.63%
CRGNet	full	80.62%	53.58%	69.06%	64.70%	29.43%	63.22%	43.41%	74.28%	0.00%	84.55%
DBFNet	full	90.35%	78.22%	82.45%	84.84%	63.12%	78.47%	**77.13%**	84.54%	64.30%	90.93%
Pixel-Net	full	**92.28%**	**79.08%**	**86.54%**	**86.36%**	**64.83%**	**82.54%**	64.67%	**87.67%**	64.62%	**95.39%**
SVM	point	57.95%	37.05%	38.68%	56.04%	0.00%	4.11%	**59.30%**	55.41%	0.00%	82.83%
RF	point	62.37%	40.14%	38.99%	46.46%	12.12%	36.90%	46.32%	56.32%	0.04%	**84.01%**
LinkNet	point	47.15%	22.18%	35.82%	26.67%	28.21%	32.99%	0.10%	32.22%	0.79%	20.63%
Deeplabv3+	point	61.60%	34.69%	37.87%	50.86%	34.22%	39.29%	5.29%	56.52%	0.69%	52.78%
HRNetv2	point	68.70%	44.56%	46.70%	63.97%	38.60%	43.82%	31.81%	62.69%	1.42%	67.45%
CRGNet	point	51.23%	23.22%	36.80%	29.56%	1.36%	26.83%	0.00%	40.55%	0.00%	50.69%
DBFNet	point	69.88%	42.14%	48.76%	61.66%	19.22%	43.40%	18.29%	69.30%	**3.24%**	73.24%
Pixel-Net	point	**76.53%**	**50.92%**	**59.57%**	**67.80%**	**46.07%**	**57.52%**	34.16%	**70.44%**	2.60%	69.19%

different layers. Deeplabv3+ captures dilated convolution and uses an encoder–decoder architecture to handle segmentation tasks. HRNetv2 uses a multibranch design and step-by-step fusion mechanism to process high-resolution images effectively, preserving both detailed and global information to improve segmentation results. CRGNet expands the annotation region by iteratively selecting unlabeled pixels with high confidence. DBFNet incorporates the bilateral filtering technique into deeply learned representations, achieving a consistent prediction in the smooth region.

To guarantee fairness and impartiality in the comparison, several measures were implemented throughout the training process. First, for the machine learning approach, each sample contains spectral information from a single point sample as well as texture features computed using the Gabor operator. It is important to note that these samples were not reused during the training phase. On the other hand, for the semantic segmentation methods, the training samples consisted of 256 × 256 image blocks comprising three spectral bands. Only a minimal percentage of pixels (0.01%) within these blocks contained category information. To prevent any overlap or reuse of samples, they were cropped with a step size of 256. This meticulous method ensured that each point sample was distinct and not reused throughout the training process. By following these procedures, the samples were determined not to have been overlapped or reused, thus maintaining the fairness of the assessment and comparison process.

Tables IV and V compare the accuracy of machine learning and semantic segmentation methods on the Vaihingen dataset and Zurich Summer dataset, respectively, highlighting the highest values in bold. The results show that semantic segmentation performs better than machine learning on the

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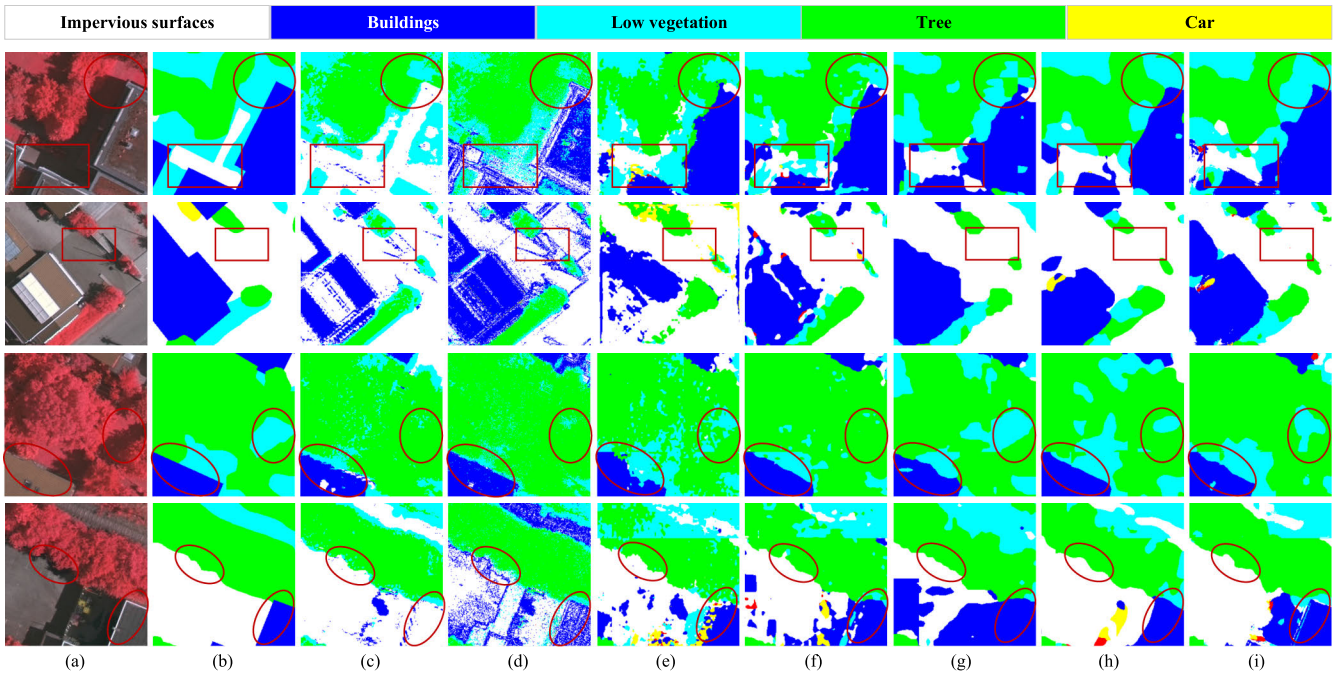


Fig. 10. Example of weakly supervised semantic segmentation results produced by different methods on the Vaihingen dataset. (a) Original image. (b) Ground truth. (c) SVM. (d) RF. (e) Deeplabv3+. (f) HRNetv2. (g) CRGNet. (h) DBFNet. (i) Pixel-Net.

larger Vaihingen dataset, while both methods achieve similar accuracy on the smaller Zurich Summer dataset, but with machine learning suffering from overfitting. Among the semantic segmentation methods, HRNetv2 achieves better results through multiscale semantic information extraction, despite struggling with small features due to incomplete labels and noise. CRGNet is severely affected by the number of samples and has higher extraction accuracy on the Vaihingen dataset, which has more samples, but performs poorly on the summer dataset. DBFNet performs poorly on the summer dataset because it can be smoothing the features, and it achieves higher accuracy on both datasets. Pixel-Net achieves optimal accuracy on both the Vaihingen dataset and the Zurich Summer dataset by enhancing feature edges and reducing noise through the edge enhancement and multiscale blurring modules, improving OA by 1.84% and 6.65% and MIoU by 3.95% and 8.78%, compared to DBFNet.

Figs. 10 and 11 show the weakly supervised semantic segmentation classification results for the Vaihingen and Zurich Summer datasets, respectively. As can be seen, the machine learning method is heavily influenced by noise because of the pixel-by-pixel classification, the internal consistency of the features is poorly maintained, and misclassification is more serious, with the edges of the features not being accurately represented. The semantic segmentation method is less affected by noise and maintains better internal consistency because the input is a 256 × 256 size image block, so more global information and texture information can be learned when extracting features, but there is still an obvious misclassification problem due to the lack of training information. Compared to traditional semantic segmentation methods, Pixel-Net demonstrates superior segmentation performance. Specifically, Pixel-Net effectively reduces the impact of noise on the results, resulting in classification outcomes with better internal consistency, as shown in the red rectangular region. Furthermore, Pixel-Net accurately extracts edges, displaying distinct boundaries and preserving finer details, as showcased by the red elliptical region.

2) Ablation Experiments: Ablation experiments were conducted on the Vaihingen dataset to investigate the impact of the edge enhancement module and multiscale blurring module. The accuracy of the OA and MIoU scores were evaluated as performance metrics. Three sets of comparative experiments were established for this purpose. The first set, referred to as baseline, involved removing the multiscale blurring module from the Pixel-Net base network structure. In addition, the conventional 3 × 3 convolution replaced the edge enhancement module. Subsequently, the models Baseline + E and Baseline + M were created by incorporating the edge enhancement module and multiscale blurring module into the Baseline, respectively.

The experimental results are recorded in Table VI, with the highest values highlighted in bold. The results demonstrate notable improvements in the model's classification capability. Specifically, the inclusion of the edge enhancement module and the multiscale blurring module led to an increase in MIoU by 1.21% and 0.94%, respectively, as observed from the accuracy comparison. Moreover, the accuracy in extracting small features and complex edge features, such as cars and low vegetation, significantly improved. When utilizing both the edge enhancement module and the multiscale blurring module, the model exhibited enhancements in OA and MIoU by 0.87% and 2%, respectively, accompanied by improved extraction accuracy across all categories.

The extraction results in the ablation experiments are shown in Fig. 12. By comparison, it can be found that the

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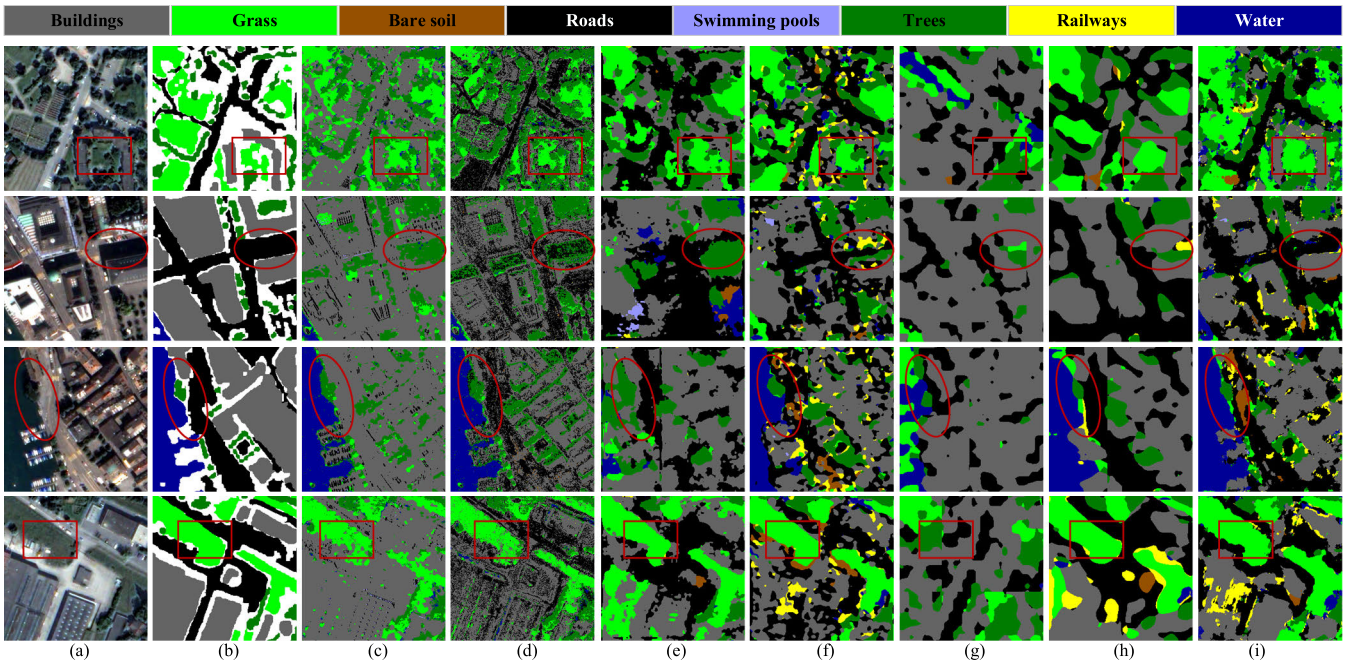


Fig. 11. Example of weakly supervised semantic segmentation results produced by different methods on the Zurich Summer dataset. (a) Original image. (b) Ground truth. (c) SVM. (d) RF. (e) Deeplabv3+. (f) HRNetv2. (g) CRGNet. (h) DBFNet. (i) Pixel-Net.

TABLE VI
INFLUENCE OF THE PROPOSED MODULES ON VAIHINGEN DATASET. E: EDGE ENHANCEMENT MODULE. M: MULTISCALE BLURRING MODULE

Method	OA	IoU	Impervious surfaces	Buildings	Low vegetation	Tree	Car
Baseline	78.81%	57.66%	68.23%	76.16%	50.65%	67.20%	26.05%
Baseline + E	79.32%	58.87%	69.24%	77.77%	**52.19%**	66.98%	28.16%
Baseline + M	78.86%	58.59%	67.93%	76.74%	52.35%	66.81%	29.12%
Pixel-Net	**79.68%**	**59.66%**	**69.59%**	**78.95%**	52.17%	**67.69%**	**29.88%**

classification results obtained by the baseline have inaccurate edge extraction of features are heavily influenced by noise, and some features, such as buildings, have cavities inside, which prevent better complete identification of the whole feature. With the addition of the edge enhancement module and multiscale blurring module, the issue of noise in the extraction results is significantly improved, as demonstrated by the red rectangular region. Moreover, Pixel-Net exhibits notably superior performance in extracting object edges compared to the baseline network, as illustrated by the red elliptical region.

3) Complexity of Pixel-Net: Our proposed algorithm extracts edge features and reduces the effect of noise through a multiscale convolution strategy to maintain accurate classification capability and robustness across the network, but this makes the model use more convolutional layers. To validate the tradeoff between performance and complexity of Pixel-Net, this article compared the trainable parameters, train time, prediction time, OA, and MIoU scores of the related methods on the Vaihingen dataset. Prediction time is the time taken by the different methods to classify the images in all test datasets, as the data reads are the same, and therefore, the read time of the data is not considered. Train time is the time spent training each epoch of the deep learning method.

Table VII shows the comparison of model performance and complexity. It can be seen that the machine learning methods take significantly more time to predict than the semantic segmentation methods, with most of the semantic segmentation methods completing classification in around 10 s, while the fastest running machine learning RF takes 395 s. The reason for this difference is that machine learning methods perform pixel-by-pixel classification, making predictions for only one pixel per run. In contrast, semantic segmentation methods perform end-to-end classification, generating category labels for each run based on the size of the input image. In this study, the size of the input image for the semantic segmentation model is set to 256 × 256, which indicates that 65 536 pixels of categories are available per run.

Compared to other semantic segmentation methods, the number of feature channels set per convolutional layer is smaller in Pixel-Net, and the Sobel convolutional layer parameters for extracting edge features are fixed and do not need to be changed during training. Therefore, Pixel-Net can better extract spectral and texture features in an image using only a small number of parameters and has faster training speed and higher classification accuracy. Specifically, the parameter of Pixel-Net is less than 1/3 of that of DBFNet, and the OA and MIoU are 1.84% and 3.95% higher, respectively.

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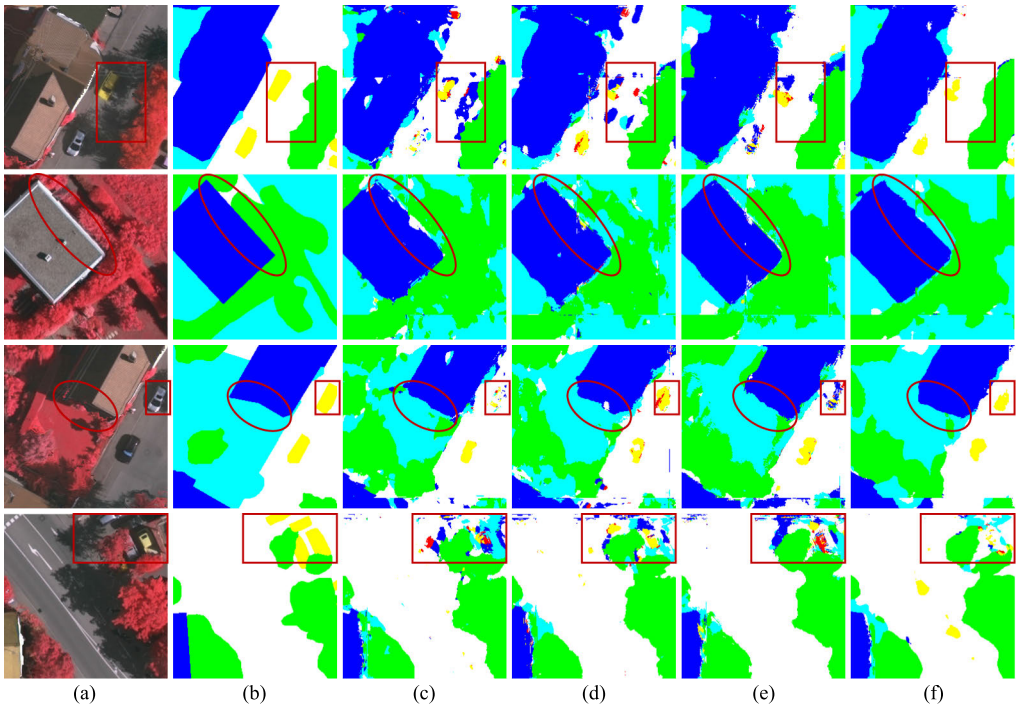


Fig. 12. Example of semantic segmentation results of ablation experiments on the Vaihingen dataset. (a) Original image. (b) Ground truth. (c) Baseline. (d) Baseline + E. (e) Baseline + M. (f) Pixel-Net.

TABLE VII

PERFORMANCE, COMPLEXITY, AND EFFICIENCY OF RELATED METHODS ON THE VAIHINGEN DATASET

Method	OA	MIOU	Parameters	Train time (second/ epoch)	Predic time (second)
SVM	65.87%	39.39%	-	-	13296
RF	69.37%	43.15%	-	-	395
LinkNet	73.16%	50.47%	11,542,188	48	9
Deeplabv3+	74.54%	50.89%	41,053,758	131	12
HRNetv2	77.40%	55.32%	28,554,592	298	15
CRGNet	74.99%	48.91%	29,241,192	101	11
DBFNet	77.84%	55.71%	11,412,144	70	11
Pixel-Net	79.68%	59.66%	3,225,838	64	9

In addition, the training time of PixelNet is only longer than that of LinkNet, requiring only 64 s to train one epoch.

D. Data Augmentation in Weakly Supervised Segmentation

Remote sensing imagery often suffers from difficult access to category information in feature classification tasks, and therefore, model training is often not adequately completed, resulting in limitations in model performance. Data augmentation techniques can be used to increase the training data through methods such as geometric transformations and pixel changes, which in turn improve the generalization and robustness of the model. To verify the effectiveness of different data augmentation methods in weakly supervised semantic segmentation, distinct data enhancement methods were employed on the Vaihingen dataset to augment the training samples. Among the data enhancement methods used are reducing the clipping step, rotation, blurring, and adding noise. The rotation refers to the random angle rotation of the image, while the added noise is salt-and-pepper noise. The blurring is achieved by smoothing the image with a mean filter. These methods expand the original data to twice its size.

Table VIII shows the changes in model accuracy for the three data enhancement methods of rotation, blurring, and adding pretzel noise used alone and in combination. It can be seen that the different data enhancement methods increase the sample size of the training set, but the OA of the model decreases. However, when rotation was used together with the other methods, the MIoU was lower than before data enhancement. When using blurring and adding pretzel noise together as data enhancement methods, the OA of the model only decreased by 0.1%, while MIoU increased by 1.2%, which indicates that the model is more capable of recognizing different features, especially for small features such as cars. Consequently, it can be deduced that blurring and pretzel noise addition enhance the learning capacity of weakly supervised semantic segmentation models for various surface types featuring different scales.

Table IX shows the change in performance of the model after reducing the clipping step size. It can be found that the number of samples in the model training dataset increases significantly with the reduction of the clipping step size. At the same time, both the OA and MIoU of the model showed an increasing trend as the clipping step size decreased. Considering the number of samples and the model performance, the clipping step size is set to 128.

Combining the above experimental results, it becomes apparent that in weakly supervised semantic segmentation utilizing point labels, diminishing the clipping step can substantially enhance the classification performance of the

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TABLE VIII
EFFECT OF DIFFERENT DATA ENHANCEMENT METHODS IN WEAK SUPERVISION ON THE VAIHINGEN DATASET

Rotate	Blur	Noise	OA	MIoU	Impervious surfaces	Buildings	Low vegetation	Tree	Car
√	√	√	77.78%	56.95%	66.52%	74.69%	50.51%	66.21%	26.84%
	√	√	79.42%	60.74%	68.92%	77.62%	52.19%	67.48%	37.50%
√	√		77.94%	56.66%	66.78%	75.15%	49.74%	66.43%	25.21%
√		√	78.52%	57.68%	68.41%	76.22%	48.82%	66.75%	28.22%
√			79.22%	60.53%	69.73%	78.28%	50.27%	66.37%	37.98%
	√		79.24%	60.04%	67.81%	77.63%	51.08%	67.83%	35.86%
		√	78.97%	59.32%	68.83%	77.75%	50.95%	66.46%	32.62%
-	-	-	79.52%	59.52%	69.19%	77.84%	51.31%	67.78%	31.49%

TABLE IX
EFFECT OF DIFFERENT CLIPPING STEP SIZES IN WEAK SUPERVISION ON THE VAIHINGEN DATASET. S128: STRIDE-128

Stride	OA	MIoU	Number of samples
stride-64	80.94%	62.79%	25589
stride-128	80.37%	61.49%	6485
stride-192	79.84%	60.61%	2992
stride-256	79.52%	59.52%	1726
s128+blur+noise	80.32%	62.10%	12970

model. In addition, when employing blurring and pretzel noise in tandem, the model's recognition capability for features of varying scales is greatly improved. The results in Table IX show that when the clipping step is set to 128 and blurring and adding pretzel noise are used at the same time, the OA score improves by 0.8% and the MIoU score improves by 2.6%.

E. Sample Expansion Evaluation

ISSE addresses the problem of lack of labeling of large amounts of data in weakly supervised semantic segmentation by iterative expansion. To validate the effectiveness of ISSE, this section shows the classification results and accuracies for each iteration on the Vaihingen dataset and the Zurich Summer dataset. Notably, the training samples used during the experiments conducted in this section are after data augmentation.

Tables X and XI show the comparison of the accuracy of the model after each iteration of ISSE on the two datasets. It can be found that the overall classification accuracy of the model shows an increasing trend as the number of iterations increases, and the improvement in model performance is most obvious in the first four iterations. This is because, as the number of iterations increases, the number of expanded sample points increases, providing more valid labeling information. Also, after the number of iterations increases, the more sample points expanded by ISSE contain error information, the more similar the training results of the model are to the model before iterations, and the model tends to stabilize. Compared to the Vaihingen dataset, the Zurich Summer dataset has fewer training data, thus resulting in a greater accuracy improvement when using ISSE for data augmentation. Specifically, after sample expansion using ISSE, OA increased by 1.68% on the Vaihingen dataset and by 2.61% on the Zurich Summer dataset. In terms of recognition accuracy for different categories, ISSE performs well in improving the shape regularity and large average footprint area of land object categories. However, it is more effective in improving the accuracy of smaller sized land object categories. This is mainly because, in the process of sample expansion, pixels in smaller land objects are more likely to be considered as edge or noise pixels, resulting in fewer expanded point annotations. On the other hand, larger sized land objects with regular shapes contain more pixels and exhibit good internal consistency, leading to a higher number of expanded point annotations. Specifically, in the Vaihingen dataset, except for the car category, the recognition accuracy of other categories has improved. In the Zurich Summer dataset, except for trees and railways, the recognition accuracy of other categories has significantly increased.

Figs. 13 and 14 show the classification results of ISSE after each iteration on both datasets. From the classification results, it is evident that the classification results obtained by the deep learning model after the training sample data enhancement can reflect the class information of the features more accurately, but there is still a small amount of noise in some places, and there is a problem of partial misclassification. With the increase of ISSE iterations, the impact of noise on the classification results diminishes progressively. Furthermore, the effect of edge extraction on the features significantly improves, as exemplified in the red rectangular region.

F. Discussion on the Classification Framework

Each of the three components in the classification framework proposed in this article is discussed in Sections III-B–III-D. To further validate the efficacy and generalizability of the proposed classification framework, experiments are conducted on the Vaihingen and Zurich Summer datasets. Specifically, alternative methods, including LinkNet, Deeplabv3+, HRNetv2, CRGNet, and DBFNet, are employed as replacements for Pixel-Net within the classification framework. The accuracy of these diverse methods is evaluated without incorporating data augmentation and ISSE.

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TABLE X
ACCURACY OF THE MODEL AFTER EACH ITERATION OF ISSUE ON THE VAIHINGEN DATASET

Iteration	OA	MIoU	Impervious surfaces	Buildings	Low vegetation	Tree	Car
itera-0	80.32%	62.10%	70.32%	79.00%	53.60%	68.03%	39.53%
itera-1	80.81%	61.61%	70.58%	78.02%	54.38%	69.82%	35.26%
itera-2	81.27%	61.95%	71.26%	78.88%	55.68%	69.99%	33.93%
itera-3	81.85%	62.05%	71.81%	79.23%	56.62%	71.12%	31.46%
itera-4	81.99%	62.24%	71.99%	79.68%	57.30%	70.91%	31.33%
itera-5	82.00%	62.12%	72.18%	80.16%	57.02%	70.72%	30.51%

TABLE XI
ACCURACY OF THE MODEL AFTER EACH ITERATION OF ISSUE ON THE ZURICH SUMMER DATASET

Iteration	OA	MIoU	Buildings	Grass	Bare soil	Roads	Swimming pools	Trees	Railways	Water
itera-0	84.65%	64.93%	69.36%	80.19%	60.13%	63.75%	64.83%	81.17%	8.54%	91.49%
itera-1	86.59%	66.84%	72.97%	80.91%	60.43%	68.29%	63.88%	84.08%	12.81%	91.37%
itera-2	86.88%	65.95%	73.95%	81.38%	57.23%	69.23%	63.43%	83.76%	7.10%	91.53%
itera-3	87.07%	66.86%	75.38%	79.76%	60.78%	70.42%	64.59%	82.58%	9.74%	91.60%
itera-4	87.18%	66.24%	74.83%	81.18%	59.06%	70.07%	62.21%	83.61%	7.11%	91.85%
itera-5	87.26%	67.05%	75.48%	80.15%	61.99%	70.78%	64.76%	83.06%	8.20%	91.96%

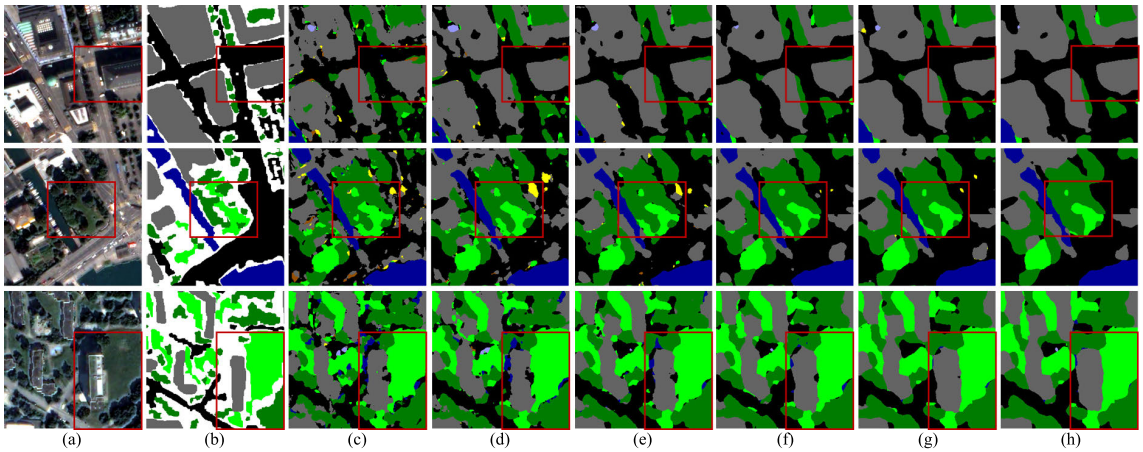


Fig. 13. Example of semantic segmentation results of each iteration on the Zurich Summer dataset. (a) Original image. (b) Ground truth. (c) Itera-0. (d) Itera-1. (e) Itera-2. (f) Itera-3. (g) Itera-4. (h) Itera-5.

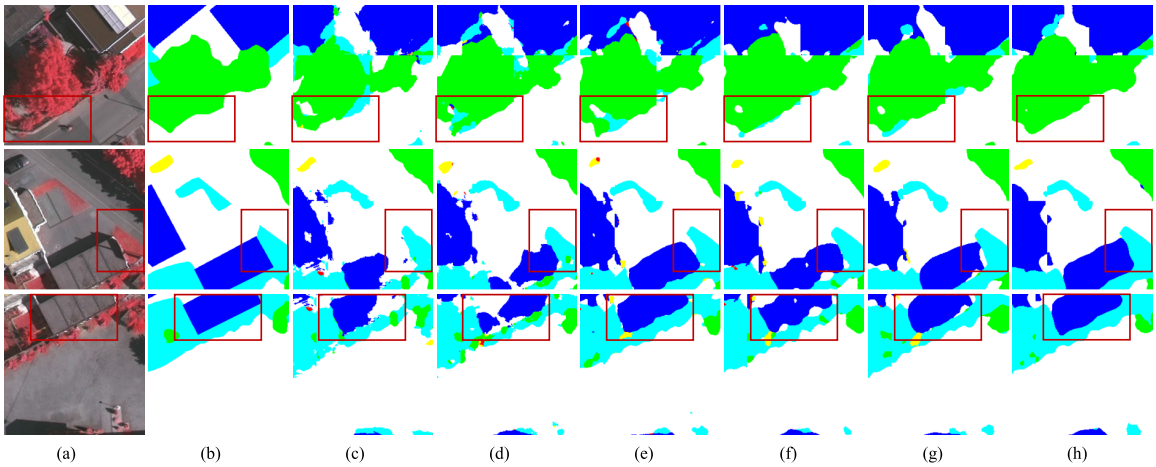


Fig. 14. Example of semantic segmentation results of each iteration on the Vaihingen dataset. (a) Original image. (b) Ground truth. (c) Itera-0. (d) Itera-1. (e) Itera-2. (f) Itera-3. (g) Itera-4. (h) Itera-5.

Table XII shows the comparison of classification accuracies of different methods. It can be seen that the accuracy of the model improves after using data enhancement for both point weakly supervision and full supervision, and the improvement

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TABLE XII
ACCURACY OF DIFFERENT MODELS IN THE WEAKLY SUPERVISED CLASSIFICATION FRAMEWORK

Vaihingen Data set						Zurich Summer Data set					
Model	Data augmentation	ISSE	Supervision	OA	MIoU	Model	Data augmentation	ISSE	Supervision	OA	MIoU
Deeplabv3+	-	-	full	81.82%	64.91%	Deeplabv3+	-	-	full	82.76%	61.82%
Deeplabv3+	√	-	full	84.27%	70.83%	Deeplabv3+	√	-	full	92.51%	80.62%
HRNetv2	-	-	full	83.87%	70.41%	HRNetv2	-	-	full	90.84%	75.81%
HRNetv2	√	-	full	**85.32%**	**73.17%**	HRNetv2	√	-	full	94.27%	83.75%
CRGNet	-	-	full	79.18%	59.45%	CRGNet	-	-	full	80.62%	53.58%
CRGNet	√	-	full	80.80%	62.43%	CRGNet	√	-	full	84.40%	62.57%
DBFNet	-	-	full	82.61%	65.53%	DBFNet	-	-	full	90.35%	78.22%
DBFNet	√	-	full	83.66%	69.51%	DBFNet	√	-	full	92.51%	81.25%
Pixel-Net	-	-	full	83.70%	69.79%	Pixel-Net	-	-	full	92.28%	79.08%
Pixel-Net	√	-	full	84.26%	71.64%	Pixel-Net	√	-	full	**94.57%**	**85.36%**
Deeplabv3+	-	-	point	74.54%	50.89%	Deeplabv3+	-	-	point	61.60%	34.69%
Deeplabv3+	√	-	point	78.53%	57.75%	Deeplabv3+	√	-	point	77.06%	56.59%
Deeplabv3+	√	√	point	80.40%	55.61%	Deeplabv3+	√	√	point	83.27%	62.57%
HRNetv2	-	-	point	77.40%	55.32%	HRNetv2	-	-	point	68.70%	44.56%
HRNetv2	√	-	point	78.93%	58.56%	HRNetv2	√	-	point	77.93%	56.89%
HRNetv2	√	√	point	80.84%	57.15%	HRNetv2	√	√	point	84.42%	63.45%
CRGNet	-	-	point	74.99%	48.91%	CRGNet	-	-	point	51.23%	23.22%
CRGNet	√	-	point	77.19%	51.92%	CRGNet	√	-	point	58.48%	33.16%
CRGNet	√	√	point	78.93%	52.72%	CRGNet	√	√	point	61.49%	30.06%
DBFNet	-	-	point	77.84%	55.71%	DBFNet	-	-	point	69.88%	42.14%
DBFNet	√	-	point	78.40%	57.05%	DBFNet	√	-	point	79.47%	59.54%
DBFNet	√	√	point	79.75%	55.98%	DBFNet	√	√	point	80.26%	59.89%
Pixel-Net	-	-	point	79.68%	59.66%	Pixel-Net	-	-	point	76.53%	50.92%
Pixel-Net	√	-	point	80.32%	62.10%	Pixel-Net	√	-	point	84.65%	64.93%
Pixel-Net	√	√	point	**82.00%**	**62.12%**	Pixel-Net	√	√	point	**87.26%**	**67.05%**

of the model accuracy is more obvious in the case of using point weakly supervision, which is because point weakly supervision is more lacking in useful labeling information, and data enhancement can significantly improve this problem. Similarly, the use of ISSE significantly improves the classification accuracy of all semantic segmentation models under point weakly supervision. Overall, the use of data augmentation and ISSE can significantly improve the problems of missing labeling information and a small number of samples in weakly supervised semantic segmentation and can significantly improve the classification accuracy. Furthermore, the classification framework proposed in this article demonstrates significant enhancements across various semantic segmentation models, highlighting its superior generalizability. This characteristic is advantageous for the future application of the method in large-scale mapping of remote sensing images.

V. CONCLUSION

In this article, a novel framework for weakly supervised remote sensing image classification is proposed, focusing on the influence of sample point quantity in semantic labels and data augmentation techniques on weakly supervised deep learning. Within the classification framework, a lightweight weakly supervised network (Pixel-Net) is designed, along with an iterative superpixel-based sample expansion method. These components effectively utilize fragmented category information present in point annotated semantic labels while mitigating the impact of noise on the final results. To assess the effectiveness of the proposed methods, comprehensive experiments are conducted using the Vaihingen and Zurich Summer datasets. The obtained numerical and visual results demonstrate that the proposed approach achieves higher classification accuracy and superior classification outcomes compared to conventional semantic segmentation models when employing point annotated semantic labels.

Although the method in this article achieves a higher accuracy of weakly supervised classification based on point annotations, there are still some limitations: 1) it is not effective in recognizing small features such as roads and cars when using point annotations and 2) texture information is not considered when calculating loss features. In future research, the ways to consider more image shape information in the loss function will be explored to enhance the model's ability to extract edge features and small features. In addition, the potential applications of the proposed method in the field of large-scale mapping will be thoroughly investigated.

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